

2. If the stored counter offset is greater than or equal to 2^{31} , a key refresh is required.
3. Otherwise, a key refresh is allowed, but not required.

4.22.4. Protocol Operation

4.22.4.1. Encryption Procedure

When sending a Check-In message, the server SHALL encrypt the application payload (i.e. [Check-In Counter](#) and application data provided) following these steps. If the encryption procedure fails to generate the Check-In message, the server SHALL NOT send the associated Check-In message.

1. The server SHALL generate the nonce as described in [Section 4.22.3.1, “Nonce Generation”](#).
 - a. If the generation failed, the encryption procedure SHALL return FAILURE.
 - b. If the generation succeeded, the server SHALL continue from step 2.
2. The server SHALL encrypt the plaintext as described in [Section 4.22.3.2, “Authenticated encryption”](#).
 - a. The plaintext SHALL be a concatenation of the following, in order :
 - i. The [Check-In Counter](#) value used in the encryption process and
 - ii. the Application Data, if used by the use case.
 - b. If the encryption fails, the encryption procedure SHALL return FAILURE.
 - c. If the encryption succeeds, the server SHALL continue from step 3.
3. The server SHALL create the payload string by concatenating the nonce and the output of the AEAD function.

payload = nonce || Ciphertext

Check-In Protocol Encryption Diagram